CSE 423 Quiz -3

Set: A

Mark: 20+5(bonus), Time: 20 minutes

Name: ID: Section:

1. For a point P (10, 20, 40), calculate the projected point P' using Cavalier Projection, orientation angle is 30° for projection plane Z = 5 10

$\alpha = 45°,\ \beta = 30°$

$$P' = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & \frac{\cos 30^\circ}{\tan 45^\circ} & 0 \\ 0 & 1 & \frac{\sin 30^\circ}{\tan 45^\circ} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -5 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 20 \\ 40 \\ 1 \end{bmatrix}$$

Translate back

Formulae for XY Plane

Translate Z=5 plane to Z=0/XY Plane

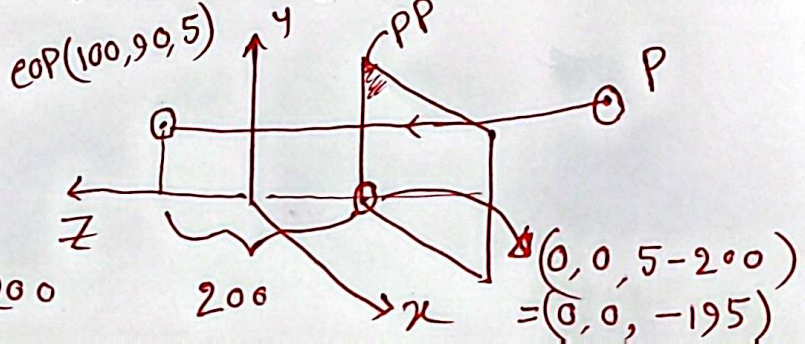
$$P' = \begin{bmatrix} 40.31 \\ 37.5 \\ 5 \\ 1 \end{bmatrix}$$

2. For COP at (100, 90, 5), calculate the projected point P' for a point P (30, 50, -245) where the Projection Plane (PP) is 200 units away from COP along Z axis 10

$COP = (100, 90, 5)$

$(0, 0, Z_p) = (0, 0, -195)$

(-) $q_x = 100,\ q_y = 90,\ q_z = 200$



$$\begin{bmatrix} 1 & 0 & -\frac{1}{2} & -97.5 \\ 0 & 1 & -\frac{9}{20} & -87.75 \\ 0 & 0 & \frac{39}{40} & -4.875 \\ 0 & 0 & -\frac{1}{200} & \frac{1}{40} \end{bmatrix} \times \begin{bmatrix} 30 \\ 50 \\ -245 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 55 \\ 72.5 \\ -243.7 \\ 1.25 \end{bmatrix} \longleftrightarrow \begin{bmatrix} 44 \\ 58 \\ -195 \\ 1 \end{bmatrix}$$

3. Write two differences between parallel and perspective projection. 5

Slide

CSE 423 Quiz -3

Set: B

Mark: 20+5(bonus), Time: 20 minutes

Name: ID: Section:

1. For a point P (10, 20, 40), calculate the projected point P' using Cabinet Projection, orientation angle is 30° for projection plane Z = -5 10

$\alpha = 63.4°$, $\beta = 30°$

$$P' = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -5 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & \frac{\cos 30°}{\tan 63.4°} & 0 \\ 0 & 1 & \frac{\sin 30°}{\tan 63.4°} & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 5 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 10 \\ 20 \\ 40 \\ 1 \end{bmatrix}$$

Translate back

formulae for XY Plane

Translate Z = -5 plane to Z = 0 / XY

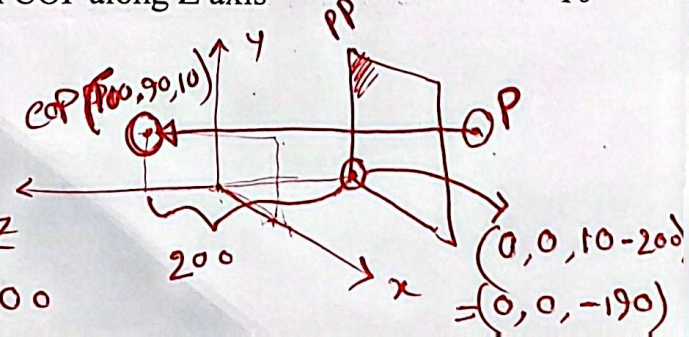
$$P' = \begin{bmatrix} 29.48 \\ 31.25 \\ -5 \\ 1 \end{bmatrix}$$

2. For COP at (100, 90, 10), calculate the projected point P' for a point P (30, 50, -240) where the Projection Plane (PP) is 200 units away from COP along Z axis 10

COP = (100, 90, 10)

$(0, 0, Z_p) = (0, 0, -190)$

(-) $q_x = 100, q_y = 90, q_z = 200$



$$\begin{bmatrix} 1 & 0 & -\frac{1}{2} & -95 \\ 0 & 1 & -\frac{9}{20} & -85.5 \\ 0 & 0 & \frac{19}{20} & -9.5 \\ 0 & 0 & \frac{-1}{200} & \frac{1}{20} \end{bmatrix} \times \begin{bmatrix} 30 \\ 50 \\ -240 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 245 \\ 243.5 \\ -237.5 \\ 1.25 \end{bmatrix} \leftrightarrow \begin{bmatrix} 196 \\ 194.8 \\ -190 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 55 \\ 72.5 \\ -237.5 \\ 1.25 \end{bmatrix} \leftrightarrow \begin{bmatrix} 44 \\ 58 \\ -190 \\ 1 \end{bmatrix}$$

3. Explain what is vanishing point and foreshortening? 5

Slide

CSE 423 Quiz -3

Set: C

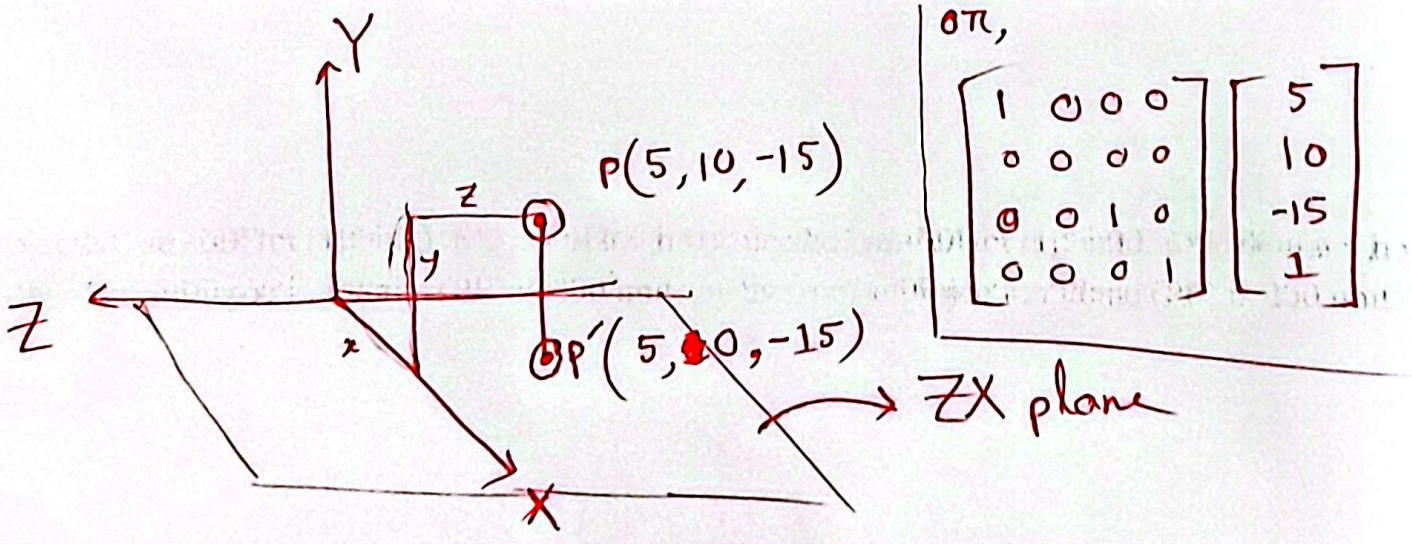
Mark: 20+5(bonus), Time: 20 minutes

Name: ID: Section:

1. For a point P (10, 20, 40), calculate the projected point P' using Cavalier Projection, orientation angle is 30° for projection plane Z = 5 10

2. For COP at (100, 90, 5), calculate the projected point P' for a point P (30, 50, -245) where the Projection Plane (PP) is 200 units away from COP along Z axis 10

3. Calculate the orthographic projected point P' of a point P (5, 10, -15) on ZX plane 5



$$\text{OT} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 10 \\ -15 \\ 1 \end{bmatrix}$$

CSE 423 Quiz -3

Set: D

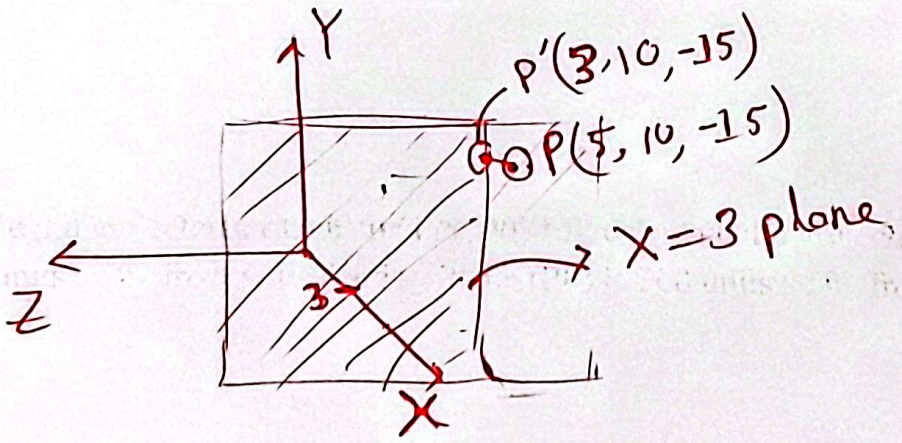
Mark: 20+5(bonus), Time: 20 minutes

Name: ID: Section:

1. For a point P (10, 20, 40), calculate the projected point P' using Cabinet Projection, orientation angle is 30° for projection plane Z = -5 10

2. For COP at (100, 90, 10), calculate the projected point P' for a point P (30, 50, -240) where the Projection Plane (PP) is 200 units away from COP along Z axis 10

3. Calculate the orthographic projected point P' of a point P (5, 10, -15) on X=3 plane 5



OR,

$$\begin{bmatrix} 0 & 0 & 0 & 3 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 5 \\ 10 \\ -15 \\ 1 \end{bmatrix}$$